

NUMERICAL ANALYSIS

GUIDED LAB 2

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Note, all the files associated with Guided Lab 2 are checked into the :

<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/GuidedLab2>

Exercise 1. We've added the code files to github at the following links: `gram_schmidt.m` `problem1.m`

(a) We went about completing this problem by writing the following code:

```
function Q = gram_schmidt( X )
    % Q = gram_schmidt( X )
    % X - is an m by n matrix
    % Q - is an m by r matrix with 1 <= r <= n that is linearly independent
    % computes the Gram-Schmidt left decomposition matrix Q
    % By Rob Tompkins, 20260215
    matrix_size = size(X);
    m = matrix_size(1,1);
    n = matrix_size(1,2);
    if X == zeros(m,n)
        error('There does not exist any type of basis for the zero vector space.');
```

```
    elseif n == 1
        Q = X(1:m,1)/norm(X(1:m,1));
    else
        if rank(X) ~= n
            X = basis_col(X);
        end
        matrix_size = size(X);
        m = matrix_size(1,1);
        n = matrix_size(1,2);
        Q = X(1:m,1)/norm(X(1:m,1));
        for i = 2:n
            u = X(1:m,i);
            v = zeros(m,1);
            for j = 1:(i - 1)
                v = v - dot(u,Q(1:m,j))*Q(1:m,j);
            end
            v_ = u + v;
            Q(1:m,i) = v_/norm(v_);
        end
    end
end
```

(b) We have two means by seeing that $X = \begin{bmatrix} 1 & 1 & 1; & 0 & 1 & 1; & 0 & 0 & 1 \end{bmatrix}$ the Gram-Schmidt algorithm should yield I_3 , the identity matrix on $\mathbb{R}^3$. The first way to notice is that it is an upper triangular matrix with no repeated columns. The other is that if we run the algorithm on the matrix X in matlab (see problem1.m for the matlab output) we get

$$Q1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(c) Similar to problem (b) it is easy to see that $X = \begin{bmatrix} 1 & 1 & 1; & 1 & 1 & 0; & 1 & 0 & 0 \end{bmatrix}$ has determinant -1, and we can see that $\text{span}(X) = \text{span}\{e_1, e_2, e_3\}$ where $e_i \in \mathbb{R}^3$ is the vector that is all zeros except for 1 in the i^{th} -place. Furthermore, we ran our Gram-Schmidt process on X and calculated the norm over the column vectors and we got 1.

(d) We ran the Gram-Schmidt algorithm on the matrix $X = \begin{bmatrix} 2 & -1 & 0; & 0 & -1 & 2; & -1 & 0 & 0; -12- \\ 1 \end{bmatrix}$ to get the following Q matrix

$$Q = \begin{bmatrix} 0.816496580927726 & 0.182574185835056 & 0.000000000000000 \\ 0 & -0.547722557505166 & 0.816496580927726 \\ -0.408248290463863 & -0.365148371670111 & -0.408248290463863 \\ -0.408248290463863 & 0.730296743340221 & 0.408248290463863 \end{bmatrix}.$$

If you look at the problem1.m file that we've linked to you'll see that each of the columns has length 1.

(e) It's quite impossible for Q to be orthogonal because $Q^T Q \in M_{3 \times 3}(\mathbb{R})$ and $Q Q^T \in M_{4 \times 4}(\mathbb{R})$, and $Q Q^T$ has non-zero entries off the diagonal.

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3.

4.